CS 556: Mathematics for Machine Learning
Assignment-8

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Question 1

Use the limit definition of the derivative to exactly evaluate the derivative:

1. $f(x) = \sqrt{x+4}$
2. $f(x) = \frac{3}{x}$

Solution

The derivative of a function using the limit definition is given by:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{1}$$

(1) $f(x) = \sqrt{x+4}$
Applying the limit definition:

$$f'(x) = \lim_{h \to 0} \frac{\sqrt{(x+h)+4} - \sqrt{x+4}}{h} \tag{2}$$

Multiplying by the conjugate:

$$f'(x) = \lim_{h \to 0} \frac{(\sqrt{x+h+4} - \sqrt{x+4})(\sqrt{x+h+4} + \sqrt{x+4})}{h(\sqrt{x+h+4} + \sqrt{x+4})} \tag{3}$$

Since $(a-b)(a+b) = a^2 - b^2$, we simplify:

$$f'(x) = \lim_{h \to 0} \frac{(x+h+4) - (x+4)}{h(\sqrt{x+h+4} + \sqrt{x+4})} \tag{4}$$

$$= \lim_{h \to 0} \frac{h}{h(\sqrt{x+h+4} + \sqrt{x+4})} \tag{5}$$

Canceling h:

$$= \lim_{h \to 0} \frac{1}{\sqrt{x+h+4} + \sqrt{x+4}} \tag{6}$$

Taking the limit:

$$f'(x) = \frac{1}{2\sqrt{x+4}} \tag{7}$$

(2) $f(x) = \frac{3}{x}$
Rewriting:

$$f(x) = 3x^{-1} \tag{8}$$

Using the power rule:

$$f'(x) = -3x^{-2} = -\frac{3}{x^2} \tag{9}$$

Thus, the derivative is:

$$f'(x) = -\frac{3}{x^2} \tag{10}$$

Question 2

Find the derivatives of the following functions:

1. $f(x) = 3x^3 - \frac{4}{x^2}$
2. $f(x) = (4 - x^2)^3$
3. $f(x) = e^{\sin(x)}$
4. $f(x) = \ln(x+2)$
5. $f(x) = x^2 \cos(x) + x \tan(x)$
6. $f(x) = \sqrt{3x^2 + 2}$
7. $f(x) = \frac{x}{4} \sin^{-1}(x)$
8. $x^2 y = (y+2) + xy \sin(x)$

Solution

1. $f(x) = 3x^3 - \frac{4}{x^2}$
Solution: Differentiate each term separately:

$$\frac{d}{dx}(3x^3) = 9x^2, \quad \frac{d}{dx}\left(-\frac{4}{x^2}\right) = \frac{8}{x^3}$$

Final result:

$$f'(x) = 9x^2 + \frac{8}{x^3}$$

2. $f(x) = (4 - x^2)^3$
Solution: Using the chain rule, let $u = 4 - x^2$, then:

$$\frac{d}{dx}u^3 = 3u^2 \cdot \frac{du}{dx}$$

Since $\frac{du}{dx} = -2x$, we get:

$$f'(x) = -6x(4 - x^2)^2$$

3. $f(x) = e^{\sin(x)}$
Solution: Let $u = \sin(x)$, then by chain rule:

$$\frac{d}{dx}e^u = e^u \cdot \cos(x)$$

Final result:

$$f'(x) = e^{\sin(x)}\cos(x)$$

4. $f(x) = \ln(x + 2)$
Solution: Using the derivative formula for logarithms:

$$f'(x) = \frac{1}{x + 2}$$

5. $f(x) = x^2\cos(x) + x\tan(x)$
Solution: Apply the product rule:

$$\frac{d}{dx}(x^2\cos(x)) = 2x\cos(x) - x^2\sin(x)$$

$$\frac{d}{dx}(x\tan(x)) = 1 \cdot \tan(x) + x\sec^2(x)$$

Final result:

$$f'(x) = 2x\cos(x) - x^2\sin(x) + \tan(x) + x\sec^2(x)$$

6. $f(x) = \sqrt{3x^2+2}$

Solution: Rewrite in exponent notation:

$$f(x) = (3x^2+2)^{1/2}$$

Using the chain rule:

$$f'(x) = \frac{1}{2}(3x^2+2)^{-1/2} \cdot 6x$$

Final result:

$$f'(x) = \frac{3x}{\sqrt{3x^2+2}}$$

7. $f(x) = \frac{x}{4}\sin^{-1}(x)$

Solution: Apply the product rule, let $u = \frac{x}{4}$, $v = \sin^{-1}(x)$:

$$f'(x) = u'v + uv'$$

Since $u' = \frac{1}{4}$ and $v' = \frac{1}{\sqrt{1-x^2}}$, we get:

$$f'(x) = \frac{1}{4}\sin^{-1}(x) + \frac{x}{4\sqrt{1-x^2}}$$

8. Given the implicit function:

$$x^2y = (y+2) + xy\sin(x)$$

Solution: Differentiate both sides using implicit differentiation:

$$\frac{d}{dx}(x^2y) = 2xy + x^2\frac{dy}{dx}$$

$$\frac{d}{dx}[(y+2) + xy\sin(x)] = \frac{dy}{dx} + x\sin(x) + y\cos(x) + y\sin(x)$$

Rearrange terms:

$$x^2\frac{dy}{dx} - \frac{dy}{dx} = -2xy - x\sin(x) - y\cos(x) - y\sin(x)$$

Factor $\frac{dy}{dx}$:

$$\frac{dy}{dx}(x^2-1) = -2xy - x\sin(x) - y\cos(x) - y\sin(x)$$

Final result:

$$\frac{dy}{dx} = \frac{-2xy - x\sin(x) - y\cos(x) - y\sin(x)}{x^2-1}$$

Question 3

The following questions consider the wind speeds of Hurricane Katrina, which affected New Orleans, Louisiana, in August 2005. The data are displayed in the table below:

Hours After Midnight, August 26	Wind Speed (mph)
1	45
5	75
11	100
29	115
49	145
58	175
73	155
81	125
85	95
107	35

(a) Using the table, estimate the derivative of the wind speed at hour 39. What is the physical meaning?

(b) Estimate the derivative of the wind speed at hour 83. What is the physical meaning?

Solution

To estimate the derivative of the wind speed at a given hour, we use the difference quotient formula:

$$f'(x) \approx \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

where x_1 and x_2 are the closest time points surrounding the given hour, and $f(x)$ represents the wind speed.

(a) Estimate $f'(39)$

From the table:

- At hour 29, wind speed = 115 mph
- At hour 49, wind speed = 145 mph

Using the difference quotient:

$$f'(39) \approx \frac{145 - 115}{49 - 29} = \frac{30}{20} = 1.5 \text{ mph per hour}$$

The wind speed was increasing at a rate of 1.5 mph per hour at hour 39.
(b) Estimate $f'(83)$
From the table:

- At hour 81, wind speed = 125 mph
- At hour 85, wind speed = 95 mph

Using the difference quotient:

$$f'(83) \approx \frac{95 - 125}{85 - 81} = \frac{-30}{4} = -7.5 \text{ mph per hour}$$

The wind speed was decreasing at a rate of 7.5 mph per hour at hour 83.

Question 4

The famous Regiomontanus' problem for angle maximization was proposed during the 15th century. A painting hangs on a wall with the bottom of the painting a distance a feet above eye level, and the top b feet above eye level.

What distance x (in feet) from the wall should the viewer stand to maximize the angle subtended by the painting, θ?

Solution

The famous Regiomontanus' angle maximization problem requires us to find the distance x from the wall that maximizes the angle θ subtended by a painting.

Step 1: Define the Problem

The painting is positioned between heights a and b above eye level. The viewer stands x feet away from the wall. The angles from the viewer's eye to the bottom and top of the painting are:

- $\alpha = \tan^{-1}\left(\frac{a}{x}\right)$ (angle to the bottom of the painting),
- $\beta = \tan^{-1}\left(\frac{b}{x}\right)$ (angle to the top of the painting).

Thus, the angle subtended by the painting is:

$$\theta(x) = \tan^{-1}\left(\frac{b}{x}\right) - \tan^{-1}\left(\frac{a}{x}\right).$$

We seek to maximize $\theta(x)$.

Step 2: Differentiate $\theta(x)$

Using the derivative formula:

$$\frac{d}{dx}\tan^{-1}\left(\frac{c}{x}\right) = \frac{-c}{x^2 + c^2},$$

we compute:

$$\theta'(x) = \frac{-b}{x^2 + b^2} + \frac{a}{x^2 + a^2}.$$

Setting $\theta'(x) = 0$ to find the critical point:

$$\frac{a}{x^2 + a^2} = \frac{b}{x^2 + b^2}.$$

Cross multiplying:

$$a(x^2 + b^2) = b(x^2 + a^2).$$

Expanding:

$$ax^2 + ab^2 = bx^2 + ba^2.$$

Rearrange:

$$ax^2 - bx^2 = ba^2 - ab^2.$$

Factor:

$$x^2(a - b) = ab(a - b).$$

Dividing by $(a - b)$ (assuming $a \neq b$):

$$x^2 = ab.$$

Thus, the optimal viewing distance is:

$$x = \sqrt{ab}.$$

Conclusion

To maximize the angle subtended by the painting, the viewer should stand:

$$x = \sqrt{ab}$$

feet away from the wall.

Question 5

An airline sells tickets from Tokyo to Detroit for $1200. There are 500 seats available, and a typical flight books 350 seats. For every $10 decrease in price, the airline observes an additional five seats sold.

- What should the fare be to maximize profit?
- How many passengers would be onboard?

Solution

We need to determine the optimal ticket price that maximizes the airline's profit and find the corresponding number of passengers.

Step 1: Define Variables

Let p be the price of a ticket (in dollars), and let x be the number of 10-dollar decreases in price.

Given:

- Initial price = $1200
- Initial passengers = 350
- For every $10 decrease in price, 5 more seats are sold.

Thus, the number of passengers as a function of x:

$$n(x) = 350 + 5x$$

The ticket price as a function of x:

$$p(x) = 1200 - 10x$$

Step 2: Express the Profit Function

Profit is given by:

$$\text{Profit} = \text{Revenue} - \text{Cost}$$

Since cost details are not provided, we assume the goal is to maximize revenue.

Revenue is given by:

$$R(x) = n(x) \cdot p(x) = (350 + 5x)(1200 - 10x).$$

Step 3: Maximize the Profit Function
Expanding:

$$P(x) = 350(1200) - 350(10x) + 5x(1200) - 5x(10x)$$

$$= 420000 - 3500x + 6000x - 50x^2$$

$$= 420000 + 2500x - 50x^2.$$

To maximize $P(x)$, take the derivative:

$$P'(x) = 2500 - 100x.$$

Setting $P'(x) = 0$ to find the critical point:

$$2500 - 100x = 0.$$

Solving for x:

$$x = \frac{2500}{100} = 25.$$

Step 4: Compute Optimal Price and Passenger Count

- Optimal price:
$$p(25) = 1200 - 10(25) = 950.$$
- Optimal number of passengers:
$$n(25) = 350 + 5(25) = 475.$$

Step 5: Conclusion

- The optimal fare to maximize profit is \$950.
- The number of passengers at this price is 475.